

Boosting PIDNet with Convolutional Block Attention Module (CBAM)

Forest Fire Segmentation
Challenge 2025

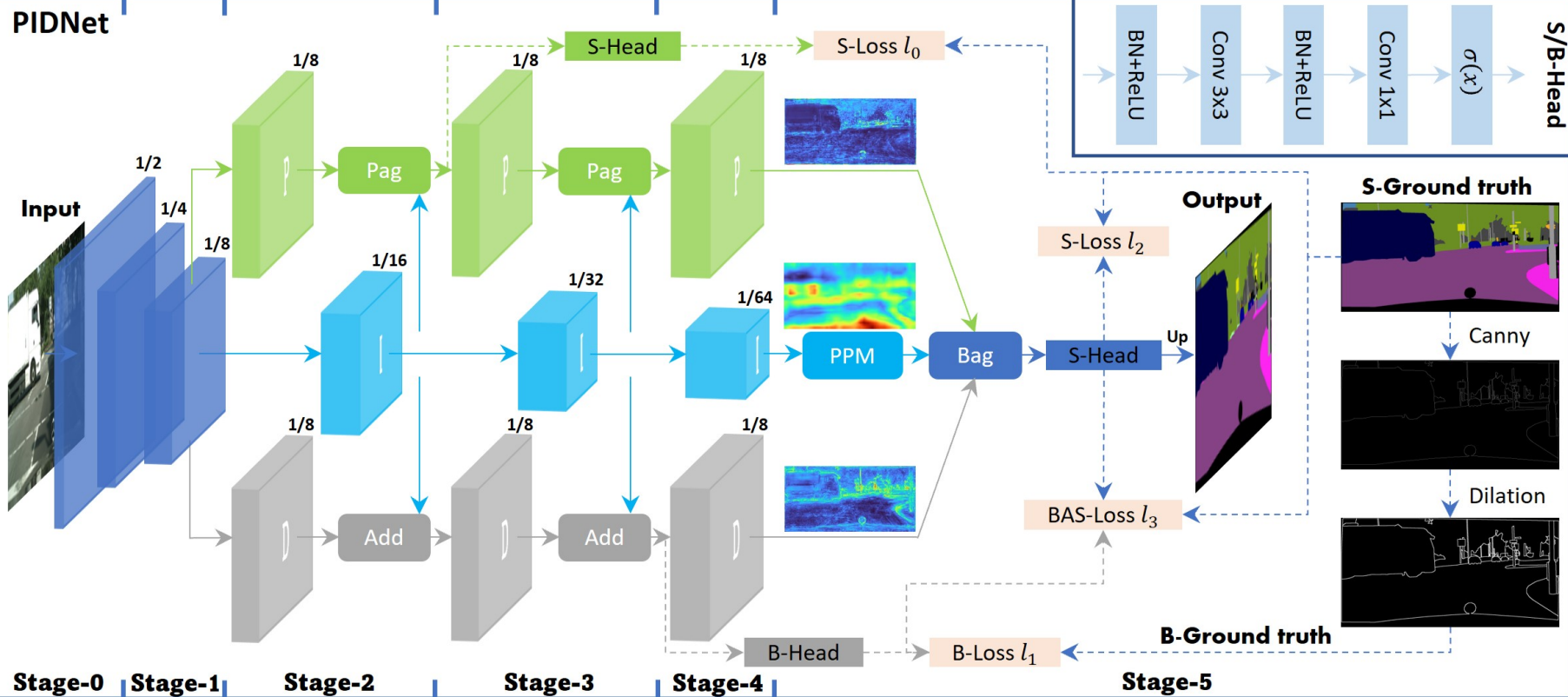
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PIDNet Structure

- It is a Real-time Semantic Segmentation Network Inspired from PID Controller
- PIDNet stands for Proportional – Integral – Derivative Network.
- The network uses three branch architecture and Resnet as the backbone.
- Specifically each branch is responsible for the following:
 - P branch: detail preservation
 - I branch: context embedding
 - D branch: boundary detection

Challenge Details

- Challenge objective: Segment fire-affected areas from aerial forest imagery.
- Base Model used: PIDNet-Medium
- Model Modification: Added CBAM to improve PIDNet's heavy downsampling



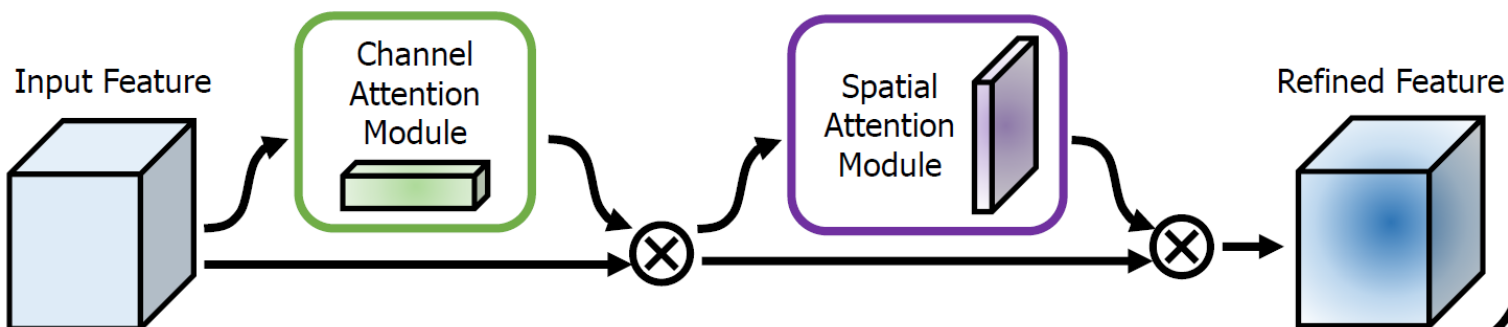
Why Touch the Backbone?

- PIDNet is fast but can miss small fire edges due to heavy downsampling.
- Attention mechanisms refine features without big cost.

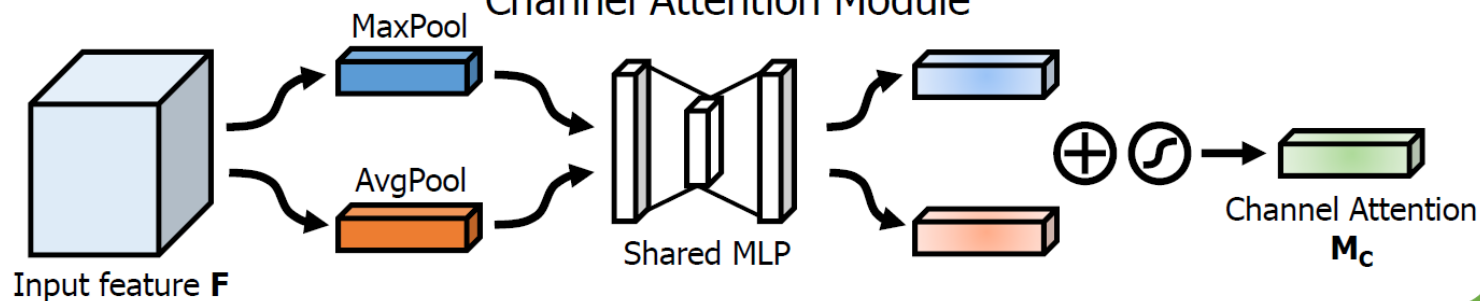
What is CBAM?

- CBAM = Channel + Spatial Attention.
- It enhances feature maps by learning to focus on “what” and “where” is important
- Channel attention → highlights what features are important (e.g., fire texture vs. background)
- Spatial attention → highlights where in the image the action is (e.g., flames at the top left)

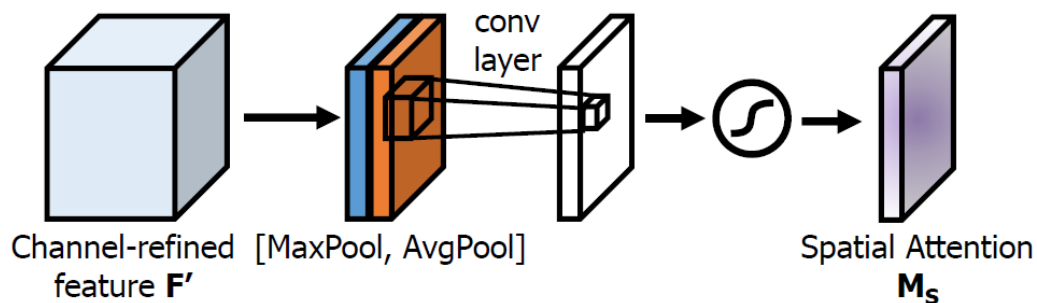
Convolutional Block Attention Module



Channel Attention Module



Spatial Attention Module



Implementaion

The code is based on the official implementation of the CBAM paper

- Link: <https://github.com/Jongchan/attention-module>
- The code has small changes so that it better matches the specific sections of the paper and the PIDNet
- It is also commented to show where exactly each section is taken from.

Base Model Changes

1st change location: Inside PAPPM fusion

Why?

- PAPPM aggregates context from multiple resolutions (e.g., pooling at 5×5 , 9×9 , 17×17), but the merged features can contain noise or unimportant info.
- By placing CBAM right after fusion, it helps:
- Reweight the important global-context regions (e.g., widespread fire zones)
- Suppress uninformative background (like distant sky or empty forest)
- It acts as a refiner of context, improving the semantic consistency of the fused features.

Base Model Changes

2nd change location: After ResNet Stage-4 (512 ch)

Why?

- This is after the deepest part of the ResNet backbone, before the features enter fusion modules like PAPPm.
- Stage-4 output (stride 1/32) carries the most abstract, semantic info but lacks spatial precision.
- By applying CBAM here:
- We help the model emphasize the most meaningful semantic channels (e.g., activations related to fire patterns)
- You refine where attention should be focused spatially — despite the low resolution

Results

Training

Seed	Model	Loss	Best mIoU	Epoch to Best	Fire IoU	Training Time
0	Baseline	0.136	0.8496	10	0.6022	8 hrs
0	CBAM	0.030	0.8824	75	0.7672	7 hrs
34	Baseline	0.151	0.8658	5	0.7342	8 hrs
34	CBAM	0.038	0.8711	30	0.7283	7 hrs
76	Baseline	0.150	0.8627	15	0.6140	8 hrs
76	CBAM	0.031	0.8871	114	0.7694	7 hrs

Table Analysis:

Overall Performance: The CBAM model consistently achieves significantly lower final training loss compared to the baseline (CBAM losses around 0.03–0.038 vs. baseline around 0.136–0.151), highlighting stronger feature extraction and optimization.

Accuracy (Best mIoU & Fire IoU): CBAM consistently surpasses baseline model in overall mIoU across all seeds, with notable improvements, especially on Seeds 0 and 76. Specifically, Fire IoU shows increases (+16.5% on Seed 0, +15.5% on Seed 76).

Convergence Speed: Although CBAM yields significantly better final results, it requires considerably more epochs (30–114 epochs) compared to the baseline (5–15 epochs). This indicates that the attention mechanism in CBAM enhances model complexity and performance but needs longer training periods.

Training Efficiency: Despite requiring more epochs, CBAM models typically trained slightly faster in wall-clock time (around 7 hours vs. 8 hours for baseline), possibly due to efficient early training phases or other implementation factors.

Results

Validation Dataset

(Settings: No flip, No multiscale, image size: 600x800)

Seed	Model	mIoU	Fire IoU	Pixel Acc.	Mean Acc.
0	Baseline	0.8496	0.7022	0.9971	0.8719
0	CBAM	0.8824	0.7672	0.9976	0.9348
34	Baseline	0.8658	0.7342	0.9974	0.8890
34	CBAM	0.8711	0.7448	0.9974	0.9155
76	Baseline	0.8627	0.7282	0.9973	0.8911
76	CBAM	0.8871	0.7765	0.9978	0.9241

Table Analysis:

CBAM consistently outperforms the baseline across all seeds on the validation set.

CBAM significantly improves Fire IoU (approximately +6% on Seed 0, +1% on Seed 34, and +5% on Seed 76).

Pixel accuracy and mean accuracy improvements are also consistent.

Results

Test Dataset

Seed	Model	mIoU	Fire IoU	Pixel Acc.	Mean Acc.
0	Baseline	0.8277	0.6572	0.9981	0.8757
0	CBAM	0.8516	0.7048	0.9983	0.9156
34	Baseline	0.8408	0.6834	0.9983	0.8814
34	CBAM	0.8041	0.6103	0.9979	0.8337
76	Baseline	0.8578	0.7173	0.9984	0.9160
76	CBAM	0.8163	0.6346	0.9980	0.8535

Table Analysis:

Test dataset results present mixed outcomes for the CBAM model. Seed 0 shows significant improvements with CBAM (+2.39% overall mIoU, +4.76% Fire IoU), validating its effectiveness. However, Seeds 34 and 76 show a noticeable decline in performance with CBAM, especially regarding the Fire IoU metric. This inconsistency suggests that CBAM, while powerful, might have sensitivity or overfitting tendencies to specific training or validation conditions, thus highlighting the necessity for additional fine-tuning and testing.

Summary & Takeaways

Training Results clearly illustrate CBAM's ability to achieve significantly lower loss values (~ 0.03 vs ~ 0.14 baseline), higher mean IoU, and notably higher IoU specifically for the critical fire class, suggesting better feature extraction and attention to important details.

Validation Dataset outcomes reinforce the advantage of CBAM, showcasing consistent improvements in both overall mean IoU (+1% to +3%) and fire-class IoU (+1% to +6%) across all seeds, reflecting superior model generalization and robustness on data similar to training.

Test Dataset performance, however, reveals mixed results: CBAM notably improves accuracy for Seed 0 (+2.39% mIoU, +4.76% fire IoU), but underperforms baseline models in Seeds 34 and 76. This variability indicates a potential sensitivity in unseen data.

In conclusion, CBAM significantly enhances PIDNet's segmentation capabilities on familiar (train and validation) datasets, but further analysis and testing are necessary to ensure consistent and robust performance on completely unseen scenarios.

Sources

- **CBAM – Convolutional Block Attention Module**
- This research can be considered as a descendant and an improvement of "Squeeze-and-Excitation Networks"
- Link: <https://arxiv.org/abs/1807.06521>
- **PIDNet – Real-time Semantic Segmentation**
- Link : <https://arxiv.org/abs/2206.02066>
- **SegNeXt: Rethinking Convolutional Attention Design for Semantic Segmentation**
- SegNeXt shows convolutional attention working best after deep feature extraction or after multi-scale fusion
- Link: <https://arxiv.org/abs/2209.08575>